EFU CHEMICAL INDUSTRY LICENCE v1.0

Experimental EFU-based Chemical & Petrochemical Framework

0. Purpose and Scope

Experimental EFU-based licensing and measurement framework for chemical industry (petrochemicals, fertilizers, plastics, pharmaceuticals).

NOT a REACH authorization, hazardous waste permit, or factory operating license replacement, but an open research protocol measuring chemical production's physical footprint in human-scale EFU metrics (feedstocks, energy, emissions, wastewater). Covers organic/inorganic synthesis, polymerization, refining.

I. Core Concepts

1. **EFU-Chemical (EFU-CHEM)**
   EFU framework application for chemical manufacturing:
   - **EFU-F_CHEM**: feedstocks (oil, gas, minerals, biomass)
   - **EFU-E_CHEM**: process energy (steam, electricity, heat)
   - **EFU-EM_CHEM**: VOCs, NOx, SOx, persistent pollutants
   - **EFU-W_CHEM**: wastewater volume + chemical oxygen demand
   - **EFU-S_CHEM**: solvents, catalysts, packaging materials
2. **Chemical Sovereignty Gap (SS_CHEM)**
   SS_CHEM = TEFU_CHEM_total – Regional Chemical Carrying Capacity (RCCC_CHEM)
   Measures if production stays within regional feedstock regeneration + waste assimilation capacity.
3. **MROI_CHEM**
   MROI_CHEM = THI_CHEM / TEFU_CHEM
   THI_CHEM = essential products supply + value added + circularity rate.

II. EFU Chemical Licence – Research Status

Documents EFU-audited chemical plants, refineries, pharma facilities. **Does NOT replace** emission permits, chemical safety reports, or product registrations. **Enables** BASF plastics vs. Yara fertilizers benchmarking.

III. Metrics and Thresholds

1. EFU Components (per tonne product)

EFU-F_CHEM (Feedstocks)
EFU-F_CHEM = feedstock_t / 300kg/person/year / product_t
Example: Ammonia = 2.8t natural gas/t NH3 → 0.0093 EFU/t NH3

EFU-E_CHEM (Energy)
EFU-E_CHEM = energy_MWh / 12.88 / product_t

EFU-EM_CHEM (Emissions)
EFU-EM_CHEM = toxics_kg / 10kg/person/year / product_t

EFU-W_CHEM (Wastewater)
EFU-W_CHEM = wastewater_m3 / 1000m3/person/year / product_t

TEFU_CHEM = 1.5×EFU-F + 1.2×EFU-E + 1.8×EFU-EM + 1.3×EFU-W + 1.1×EFU-S

2. Regional Chemical Carrying Capacity (RCCC_CHEM)

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RCCC_CHEM = Sustainable_production_Mt / Population / 2t/person/year
Example: Germany = 200Mt / 84M / 2t = 0.12 EFU_CHEM/person/year
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IV. Licence Categories (EFU-CHEM Labels)

Level	SS_CHEM	MROI_CHEM	Label	Interpretation
1	≤ 0	≥1e-2	**EFU-CHEM SOVEREIGN**	Feedstock capacity
2	0 < ≤15k EFU	≥5e-3	**EFU-CHEM BALANCED**	Manageable emissions
3	15k < ≤75k EFU	≥1e-3	**EFU-CHEM DEFICIT**	Import/emission pressure
4	>75k EFU	<1e-3	**EFU-CHEM CRITICAL**	Feedstock depletion

V. Regional RCCC_CHEM Calibrations

Country	Production (Mt/year)	Population (M)	RCCC_CHEM (EFU/person/year)
GERMANY	200	84	0.12
CHINA	1,200	1,410	0.004
USA	600	340	0.009
SAUDI ARABIA	80	37	0.12

Country	Production (Mt/year)	Population (M)	RCCC_CHEM (EFU/person/year)
EU	350	448	0.02

VI. Example: German Chemical Sector (BASF/Ludwigshafen)

Input (annual sector):

- Products: 200Mt chemicals/plastics
- Feedstocks: 150Mt oil/gas equivalents
- Energy: 120B MWh
- Toxic emissions: 5,000 tonnes
- Wastewater: 500M m³
- Workers: 500,000 staff
- Region: Germany (RCCC_CHEM = 0.12 EFU/person/year)

Calculation:

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EFU-F_CHEM = 150M/300/200M = 0.0025 EFU/t
EFU-E_CHEM = 120B/12.88/200M = 0.046 EFU/t
EFU-EM_CHEM = 5k/10/200M = 0.0000025 EFU/t

TEFU_CHEM_total ≈ 18,500,000 EFU/year
Allocated = 500k × 0.12 = 60,000 EFU/year
SS_CHEM = 18.5M – 60k = +18,440,000 EFU/year (Level 4: CRITICAL)

Result: EFU-CHEM CRITICAL
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Sub-sector Comparison: Pharmaceuticals = 0.12 EFU/t (Level 3: DEFICIT)

VII. Implementation Notes

1. **REACH extension**: TEFU_CHEM for substance authorization prioritization
2. **Carbon border tax**: EFU-F_CHEM basis for petrochemical import duties
3. **PFAS regulation**: EFU-EM_CHEM metric for persistent pollutant tracking
4. **Data sources**: Tier 1 = feedstock receipts, emission logs; Tier 2 = Cefic/ACC statistics